

Табель №1

УП-7

ИКСД-50-23

Вариант 23

$Q_i \backslash b_j$	220	150	170	200	210	U_i
350	11 11	5 5	6 4	1 9	2 6	0
	220 -	130 +	-	-	-	
320	12 6	6 6	7 7	2 2	3 3	+1
	-	20 -	170 +	130 +	-	
220	18 7	12 3	13 10	8 8	9 9	7
	- +	-	-	70 -	150 -	
60	3 0	3 0	4 0	-1 0	0 0	-2
	-	-	-	-	60	
V_k	11	5	16	1	2	

2-й - получить максимум
3-й - определить количество
 $Q_2 - b_4$ - избыточно
 $Q_1 - b_1 \geq 60$

Для Excel

$$Z = 6550$$

$$\gamma_{min} = \gamma_{31} = 7 - 18 = -11$$

$$L_{min} = (70, 20, 220) = 20$$

$Q_i \backslash b_j$	220	150	170	200	240	U_i
350	11 11	5 5	17 4	12 9	13 6	0
	200 -	150 -	- +	-	-	
320	1 6	-5 6	7 7	2 2	3 3	-10
	-	0 -	140 -	150 +	-	
220	7 7	1 3	13 10	8 8	9 9	-4
	20 +	-	-	50 -	150 -	
60	-2 0	-8 0	4 0	-1 0	0 0	-13
	-	-	-	-	60	
V_k	11	5	17	12	13	

$$Z = 6330$$

$$\gamma_{min} = \gamma_{13} = 4 - 17 = -13$$

$$L_{min} = (200, 120, 50) = 50$$

$Q_i \backslash b_j$	220	150	170	200	210	U_i
350	11 11	5 5	4 4	-1 9	13 6	0
	150 -	150 -	50 +	-	-	
320	14 6	8 0	7 7	2 2	16 3	3
	- +	-	120 -	200 -	-	
220	7 7	1 3	0 10	-5 8	9 9	-4
	70 -	-	-	-	150 -	
60	-2 0	-8 0	-9 0	-14 0	0 0	-13
	-	-	-	-	60	
V_k	11	5	4	-1	13	

$$Z = 5680$$

$$\gamma_{min} = \gamma_{24} = 6 - 14 = -8$$

$$L_{min} = (120, 150) = 120$$

$Q_i \backslash b_r$	220	150	170	200	210	U_i
350	11 11 5 5 4 4 7 9 (13 6)					0
	30 -	150	170	-	- +	
320	6 6 0 6 -1 7 2 2 (8 3)					-5
	120	-	-	200	-	
220	7 7 1 3 0 10 3 8 9 9					-4
	20 +	-	-	-	150 -	
60	-2 0 -8 0 -9 0 -6 0 0 0					-13
	-	-	-	-	60	
V_k	11	5	4	7	13	

$$Z = 4720$$

$$\gamma_{mn} = \gamma_{15} = 6 - 13 = -7$$

$$L_{mn} = (30, 150) = 30$$

$Q_i \backslash b_r$	220	150	170	200	210	U_i
350	4 11 5 5 4 4 0 9 6 6					0
	-	150 -	170	-	30 +	
320	6 6 (7 6) 6 7 2 2 (8 3)					2
	120	-	-	200	-	
220	7 7 (8 3) 7 10 3 8 9 9					3
	100 -	- +	-	-	120 -	
60	-2 0 -1 0 -2 0 -6 0 0 0					-6
	-	-	-	-	60	
V_k	4	5	4	0	6	

$$Z = 4510$$

$$\gamma_{mn} = \gamma_{32} = 3 - 8 = -5$$

$$L_{mn} = (150, 120) = 120$$

$Q_i \backslash b_r$	220	150	170	200	210	U_i
350	9 11 5 5 4 4 5 9 6 6					0
	-	30 -	170	-	150 +	
320	6 6 2 6 1 7 2 2 3 3					-3
	120	-	-	200	-	
220	7 7 3 3 2 10 3 8 4 9					-2
	100 -	120 +	-	-	-	
60	(3 0) -1 0 -2 0 -1 0 0 0					-6
	- +	-	-	-	60 -	
V_k	9	5	4	5	6	

$$Z = 3810$$

$$\gamma_{mn} = \gamma_{14} = 0 - 3 = -3$$

$$L_{mn} = (60, 30, 100) = 30$$

$q_i \backslash L_j$	220	150	120	200	210	U_i
350	6 11 -	2 5 -	4 4 120	2 9 -	6 6 180	0
320	6 6 120	2 6 -	4 7 -	2 2 200	6 3 -	0
220	7 7 20	3 3 150	5 10 -	3 8 -	7 9 -	1
60	0 0 30	-4 0 +	-2 0 -	-4 0 -	0 0 30	-6
V_k	6	2	4	2	6	

$$z = 3820$$

$$\gamma_{min} = \gamma_{2,1} = 3 - 6 = -3$$

$$L_{min} = (30, 120) = 30$$

$q_i \backslash L_j$	220	150	120	200	210	U_i
350	9 11 -	5 5 -	4 4 120	5 9 -	6 6 180	0
320	6 6 30	2 6 -	1 7 -	2 2 200	3 3 30	-3
220	7 7 20	3 3 150	2 10 -	3 8 -	4 9 -	-2
60	0 0 60	-4 0 -	-5 0 -	-4 0 -	-3 0 -	-8
V_k	9	5	4	5	6	

$$z = 3730$$

Über: $z = 3730$

Стоимость перевозок						
ai \ bk	220	150	170	200	210	950
350	11	5	4	9	6	
320	6	6	7	2	3	
220	7	3	10	8	9	
60	0	0	0	0	0	
950						
План перевозок						
	220	150	170	200	210	
350	0	90	170	0	90	
320	0	0	0	200	120	
220	160	60	0	0	0	
60	60	0	0	0	0	
Суммарная стоимость перевозок						
	0	450	680	0	540	
	0	0	0	400	360	
	1120	180	0	0	0	
	0	0	0	0	0	
						Ответ
	1120	630	680	400	900	3730

Стоимость перевозок						
ai \ bk	220	150	170	200	210	950
350	11	5	4	9	6	
320	6	6	7	2	3	
220	7	3	10	8	9	
60	0	0	0	0	0	
950						
План перевозок						
	220	150	170	200	210	
350	60	0	170	120	0	
320	110	0	0	0	210	
220	50	150	0	20	0	
60	0	0	0	60	0	
Суммарная стоимость перевозок						
	660	0	680	1080	0	
	660	0	0	0	630	
	350	450	0	160	0	
	0	0	0	0	0	
						Ответ
	1670	450	680	1240	630	4670

III

$Q_i \backslash b_j$	220	150	170	150	210
300	8	5	4	8	2
320	-	6	6	7	2
280	1	3	10	8	9
	220	60	-	-	-

$$t^F = \max(T^F) = 7$$

$$z = \min(80, 260) = 80$$

$Q_i \backslash b_j$	220	150	170	150	210
300	8	5	4	8	2
320	-	6	6	7	2
280	1	3	10	8	9
	220	60	-	-	-

$$t^F = \max(T^F) = 6$$

$$z = \min(80, 130) = 80$$

$Q_i \backslash b_j$	220	150	170	150	210
300	8	5	4	8	2
320	6	6	7	2	3
280	1	3	10	8	9
	220	60	-	-	-

$$t^F = \max(T^F) = 5$$

← Optimal